THE ARCHITECTURE OF A QUANTUM INTERNET

Mihai CATARAGA
Department of Software Engineering and Automatics, TI-254, FCIM, TUM, Chișinău, Moldova

Abstract: The current internet infrastructure is based on the classical binary system, which has fundamental limitations in computational power and cybersecurity. Recent quantum computing advancements confirm that quantum systems have a massive advantage over even supercomputers on certain tasks. For example, Google's 70-qubit Sycamore processor completed in seconds a benchmark calculation that would have required approximately 50 years on the most advanced supercomputer. In the future, larger quantum computers could pose a direct threat to modern encryption. Using Shor's algorithm, a sufficiently powerful quantum computer could render RSA encryption useless. This creates an urgent need for systems that guarantee security over communication channels. Alongside post-quantum cryptography, Quantum Key Distribution addresses this problem by using single photons or entangled photon pairs to generate shared encryption keys. Any eavesdropping attempt in this process would introduce detectable errors. Practical implementations already exist. China operates a 4,600 km integrated quantum network, and Europe's EuroQCI initiative is actively building continent-wide infrastructure. However, a full quantum internet remains impossible without solving two major engineering challenges: quantum repeaters and quantum memory, the latter not yet able to store quantum states long enough and efficiently enough for cross-continental communication. This article reviews the roadmap of quantum network infrastructure against its theoretical potential, framing quantum memory as the central bottleneck separating operational QKD from a complete quantum internet.

Key Words: *quantum computers, quantum networking, quantum key distribution, quantum internet, entanglement, quantum memory, quantum repeaters.*

Introduction

Today's internet is a global network of interconnected devices that exchange information encoded in a simple binary system. However, this system has computational and cybersecurity limitations. New quantum computing systems directly challenge those limitations and pose significant threats to today's encryption. This article explains what quantum computers are and why they threaten current encryption, presents quantum key distribution and the networks that already use it, and then examines the two engineering challenges, quantum repeaters and quantum memory, that separate these networks from a full quantum internet.

What Is a Quantum Computer?

Bits are the fundamental building blocks of current-day technology. Classical bits are always either 0 or 1, and a computer's main task is to perform calculations using this framework. That said, computationally demanding tasks far exceed the capabilities of the most advanced supercomputers. This is where quantum computers come into play. These computers use qubits, instead of normal bits. Whereas a normal bit can be either 0 or 1, qubits can be both 0 and 1, at the same time. This principle is called superposition, and it states that a particle can exist in a combination of multiple states simultaneously, until it's measured [1, 2].

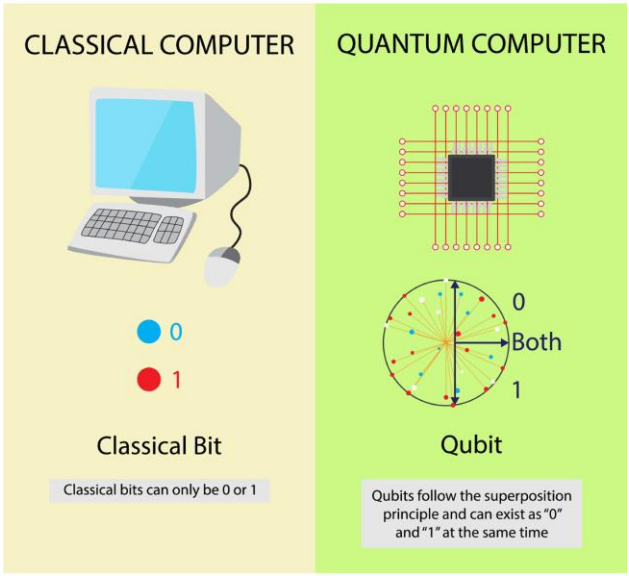


Figure 1. *Bits vs qubits. Image: Berkeley Nucleonics [3].*

Quantum computers are machines that use these qubits and can solve certain problems much faster than traditional computers, in some cases exponentially faster. This is often explained as performing many calculations in parallel, but measuring the qubits gives only one result. Instead, the speedup comes from interference. Quantum algorithms are designed so that wrong answers cancel out and the correct one becomes more likely [1, 2].

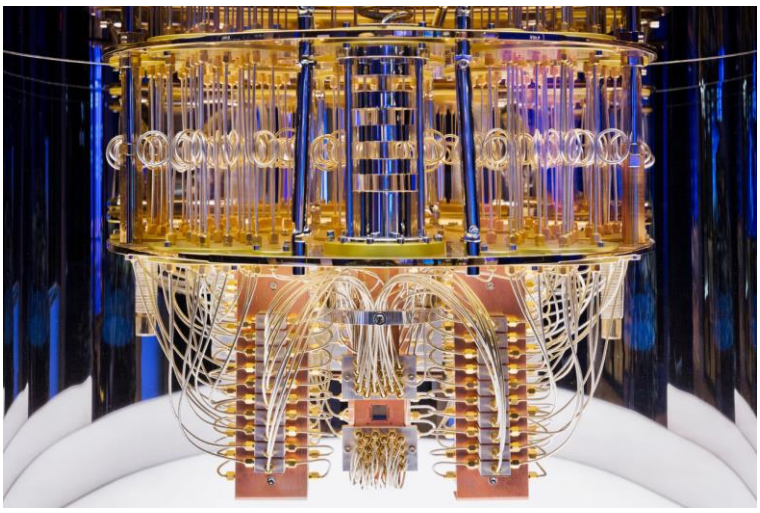


Figure 2. *IBM quantum computer. Image credit: IBM [4].*

The capabilities of these machines, however, need to be stated more precisely. For instance, in 2023 Google conducted an experiment in a controlled environment, in which they used their 70-qubit Sycamore quantum processor to perform a benchmark calculation called random circuit sampling. According to Google's experiment, it took the quantum machine mere seconds to solve the problem. For comparison, the most powerful supercomputer in the world, Frontier, would have required approximately 50 years to complete the exact same task [5, 6]. It is important to mention that this task has no practical use, but it is very hard for classical computers, which makes it a good benchmark. IBM has since reported similar results in physics simulations [7], although classical computers were able to reproduce them within weeks [8].

A further example illustrates a notable capability of quantum computers, one that at the same time poses a serious threat to modern cybersecurity. In 1994, mathematician Peter Shor developed a quantum algorithm, called Shor's algorithm, which is capable of factoring large numbers faster than any classical method [9]. This is relevant because RSA (Rivest-Shamir-Adleman) encryption, the modern-day encryption standard of internet communication, relies solely on the computational difficulty of factoring large integers. Therefore, a sufficiently powerful quantum computer running Shor's algorithm could break RSA encryption, weakening current security infrastructure. Such a machine does not exist yet, and a 2025 estimate puts the requirement at just under one million noisy qubits running for about a week [10]. Even so, the threat is already urgent: encrypted data intercepted today can be stored and decrypted once such a machine exists, a strategy known as "harvest now, decrypt later".

This urgent matter has two possible answers. The first is post-quantum cryptography, a set of new classical algorithms standardized by NIST [11]. The second is a solution where the laws of physics guarantee secure encryption. Post-quantum cryptography runs on existing hardware, but its security still rests on mathematical problems that are believed, not proven, to be hard. Physics-based methods avoid this assumption, at the cost of dedicated hardware, and they still rely on classical cryptography to authenticate the two parties. This article focuses on the second approach.

Quantum Entanglement

To understand the future of secure encryption over internet communication channels, one must first examine the complex topic of quantum entanglement. In quantum physics, two particles can become linked, or entangled with one another, such that measuring one instantly affects the other, regardless of the distance between them [1]. This, however, is not a method of information

transmission, but rather a correlation encoded at the quantum level. More generally speaking, any attempt to observe or measure a quantum state disturbs it. As the next section shows, this property makes it impossible, in principle, for an intruder who is eavesdropping to remain undetected in the network [12].

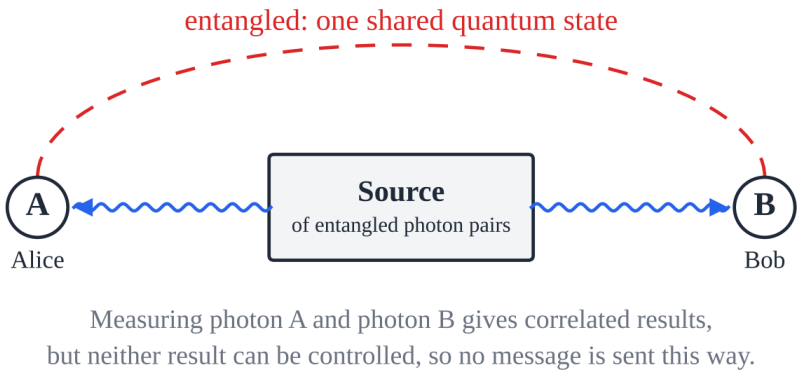


Figure 3. *A source emitting a pair of entangled photons.*

Quantum Key Distribution

Quantum Key Distribution, or QKD, is the practical application of these principles in a quantum network. QKD, similar to normal encryption, generates a shared secret key between two parties, but relies on the laws of physics instead of computational difficulty. Some protocols, such as BB84, send single photons prepared in random states [12], while others, such as E91, use entangled photon pairs [13]. Most deployed networks, including China's, use the first approach.

As mentioned before, if an eavesdropper intercepts the transmission, they are observing the quantum state of the particles, therefore disturbing them. This disturbance shows up as errors when the two parties compare a sample of their key, and the eavesdropping attempt fails [12]. Unlike RSA encryption, where the encryption keys can be broken by using powerful quantum computers, QKD doesn't necessarily make the key generation process impossible to hack, but rather makes the breach detectable, which is its specific strength. But real devices can still have weaknesses, and researchers have already hacked commercial QKD systems by attacking their detectors [14]. In such systems, a defense-in-depth approach should be used so a failure in one layer won't compromise the whole system.

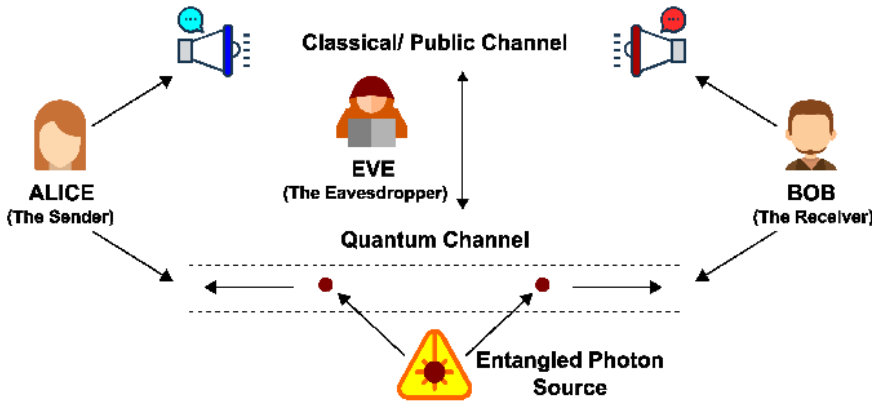


Figure 4. *The practical process of entanglement-based QKD. Source: Nurhadi and Syambas [15].*

Current Infrastructure

Technologies such as QKD are not purely theoretical. Practical implementations are already in use in China, which currently operates one of the most extensive quantum communication networks in the world, with a 4,600 km integrated space-to-ground network. It combines a 2,000 km fiber backbone between Beijing and Shanghai, made of shorter links joined by trusted relay nodes, with satellite-to-ground links [16, 17]. At each trusted relay, the key is decrypted and encrypted again, so every relay has to be physically secured and trusted. Removing this requirement is exactly what quantum repeaters are meant to do.

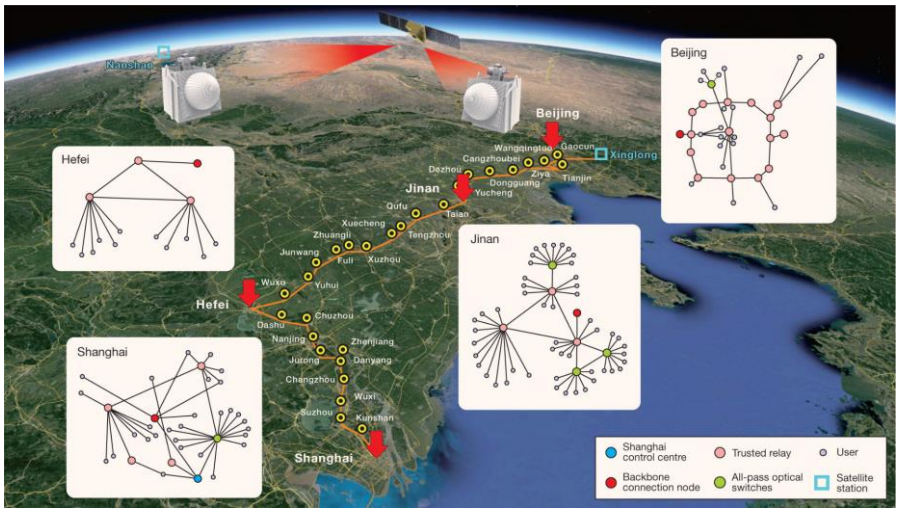


Figure 5. *The Beijing–Shanghai fiber-optic backbone. Image credit: University of Science and Technology of China, via Sci.News [18].*

Complementing the fiber network, the Micius satellite orbiting in space enabled the first intercontinental quantum-secured communication between Beijing and Vienna across 7,600 km, with the satellite itself acting as a trusted relay [19].

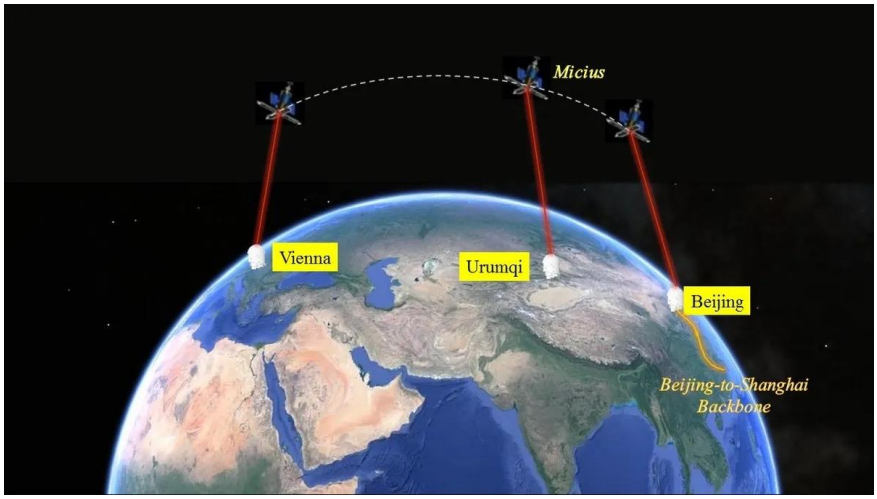


Figure 6. *The Micius satellite: QKD from Vienna to Beijing. Image credit: CAS/Pan Jianwei's team, via Space.com [20].*

Another example is in Europe, where the EuroQCI initiative is working towards establishing continent-wide quantum communication infrastructure, integrating both ground-based fiber networks and satellite links [21], similarly to China, although most of it is still at the stage of national pilot networks.

Quantum Repeaters

As is the case in all networks, no matter if they use fiber-optic or normal copper cables, the light impulse or the electric signal fades and loses intensity when transmitted over long distances. This is why classical networks use repeaters, devices that take in the weak signal, amplify it, and send it further down the network. However, in a quantum network, applying the same technology is impossible, because quantum states cannot be copied. This is known as the no-cloning theorem [22]. Therefore, it is significantly harder to find a way to retransmit the signal over the network without ever measuring the qubit directly. Quantum repeaters were first proposed in 1998 [23]. At the present moment, they exist in laboratory demonstrations and early field tests [24, 25], but no production-ready deployment exists at scale.

A quantum repeater therefore works differently. Instead of sending a photon across the full distance, the link is split into short segments, and entanglement is created across each one separately. A node in the middle then performs a joint measurement on the two photons it holds, one from each segment. This operation, called entanglement swapping, leaves the two distant endpoints entangled with each other, even though no photon traveled the whole way (Figure 7). Because each segment succeeds only at random, a node must store its half of an entangled pair until the neighboring segment confirms success. This is exactly the role of quantum memory.

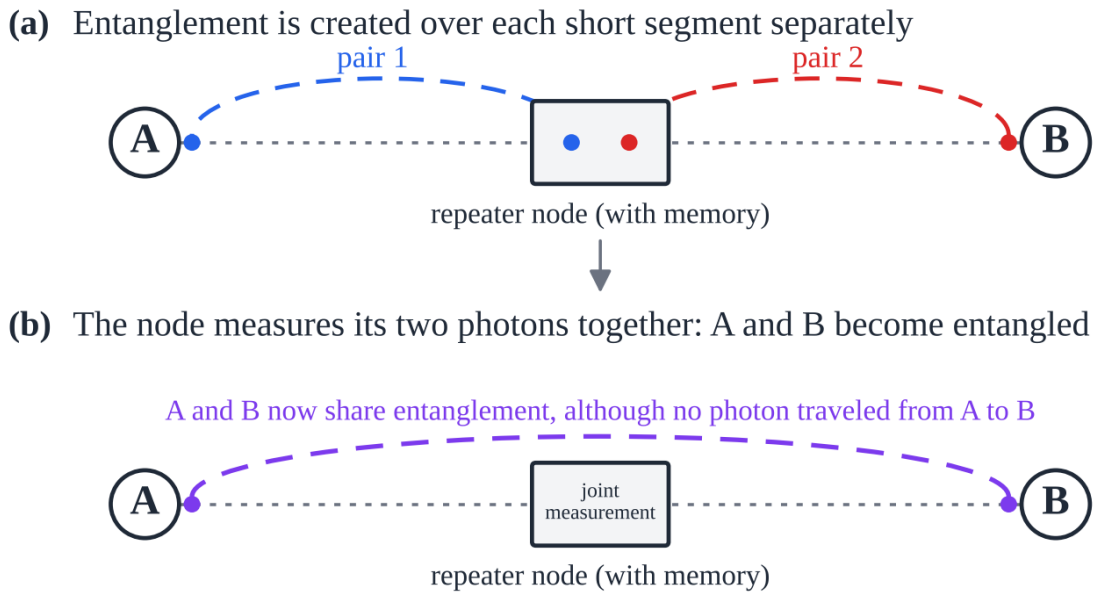


Figure 7. *Entanglement swapping at a quantum repeater node.*

Quantum Memory

This makes quantum memory the central bottleneck of a quantum network. Entangled states are fragile and degrade over time through a process called decoherence [26]. Some quantum memories can already keep their state for seconds, and one experiment measured a coherence time of six hours [27]. However, no memory yet combines long storage with the efficiency and fiber

compatibility that a network needs. Over a transatlantic distance, light in fiber needs about 30 ms just to travel one way (roughly 6,000 km at about 200,000 km/s), and failed attempts have to be repeated, so the stored states must last much longer than that. This gap is a central unsolved engineering problem of the field. Therefore, the issue of quantum memory remains an open research problem [28].

Conclusions

The quantum internet is a natural, logical extension of the current internet framework, meant to work alongside it rather than replace it. Quantum computers have already demonstrated quantum supremacy on specific benchmark tasks, and future, larger machines could pose significant threats to modern-day encryption standards. Such challenges give rise to a new model, a new way of cybersecurity using QKD networks, in which the laws of physics protect privacy. Such models already exist and operate on a large scale, which shows that this technology is of high interest and urgency. However, quantum repeaters and quantum memory remain fundamental engineering challenges, and are actively researched and studied. Such bottlenecks do not yet allow a full global quantum internet. Nevertheless, the architecture of a quantum internet is being laid out, and its first stage, trusted-repeater networks, is already in operation [29].

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